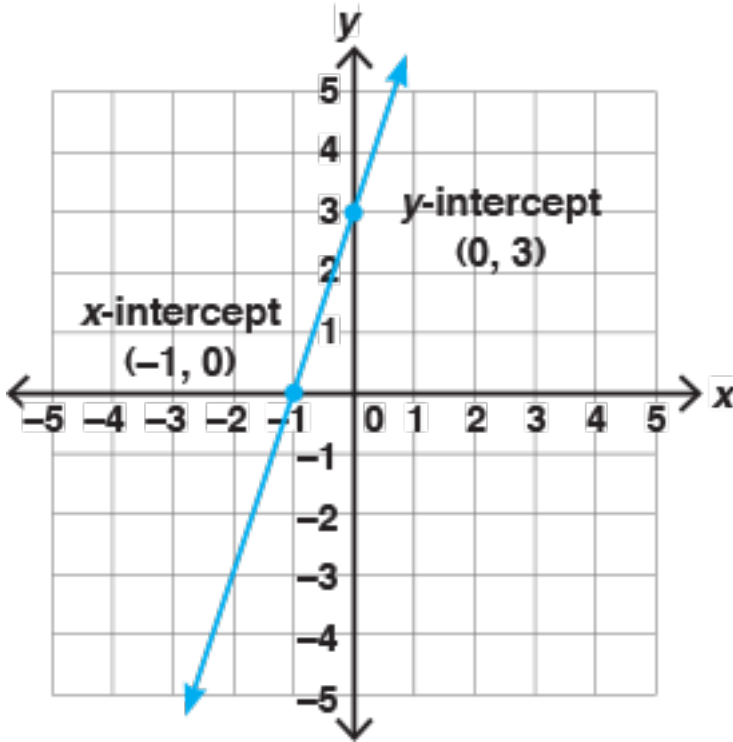


Using Intercepts

When a line crosses an axis, the points of intersection are called *intercepts*.

Look at the graph of $y = 3x + 3$ below.



When the line crosses the y -axis, the x -value is 0 and the y -value is 3.

The **y -intercept** is (0, 3).

When the line crosses the x -axis, the x -value is -1 and the y -value is 0.

The **x -intercept** is $(-1, 0)$.

You can find the intercepts of any linear equation by substituting 0 into the equation. To find the y -intercept, substitute 0 for x . To find the x -intercept, substitute 0 for y .

Example 1

[Hide](#)

Find the x - and y -intercepts of the equation $3x - 5y = 15$.

Step 1

Find the y -intercept by substituting 0 for x .

$$\begin{aligned} 3(0) - 5y &= 15 \\ -5y &= 15 \\ y &= -3 \end{aligned}$$

Step 2

Find the x -intercept by substituting 0 for y .

$$\begin{aligned} 3x - 5(0) &= 15 \\ 3x &= 15 \\ x &= 5 \end{aligned}$$

The y -intercept is **(0, -3)**. The x -intercept is **(5, 0)**.

Intercepts make it easy to graph an equation. After you find the intercepts, plot the points on a graph and draw a line to connect them.

Example 2

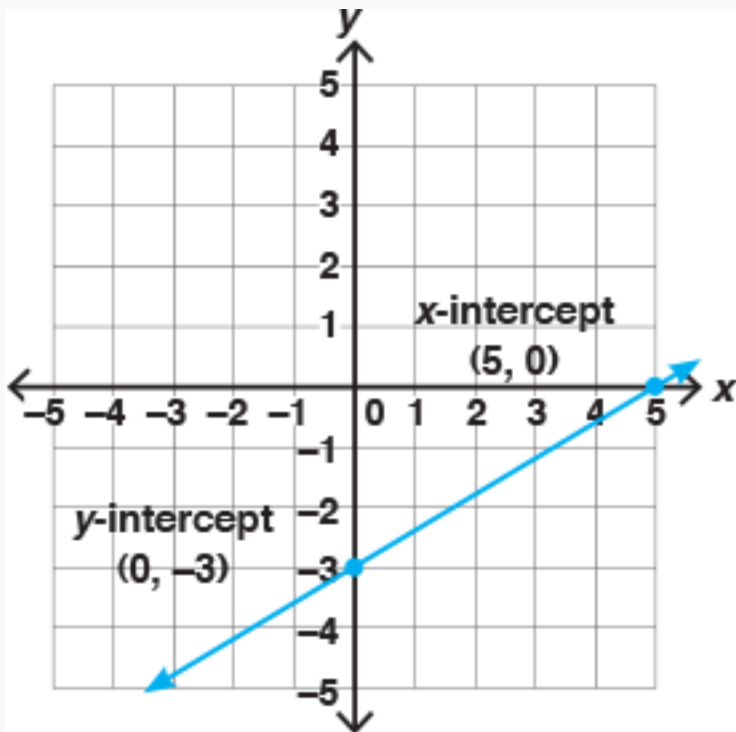
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Graph the equation $3x - 5y = 15$.

The y -intercept is (0, -3).

The x -intercept is (5, 0).

Plot the points and draw a line.



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Category

Lesson 1: Graphing a Linear Equation

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